Virtual Gold Capstone: Open-Weight Model Evaluation (Sized for Client GPU Tiers), Pipeline Hardening & Multi-Architecture Benchmark Report

Author: Anmol Sahu (Model Workstream Lead) — CMU Heinz Capstone Team
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Prepared For: Virtual Gold Inc. (Dr. Inderpal Bhandari, Alex Bhandari, Aarvin, Urte Jesina) & Capstone Engineering Team
Status: Progress & Client Update — Self-Hostable GPU Tier Matrix Defined; 6 Harness/Calibration Defects Resolved; `GLiNER` (`300M`) Zero-Shot Span Encoder & 4 Pipeline Architectures (`Configs A, B, C, D`) Integrated; 9 New Configurations Evaluated via Hosted Open-Weight APIs on 15 Development Emails (`1,198` Audited Candidate Decisions)
Repository References:

- Live Benchmark Traces (`9` Configs, `135` Email Runs): `benchmarks/2026-10-05-followup/results/free_tier_vpc_dev15_benchmark.json`
- Updated Calibration Profile (`v1.4.0`, `1,198` Records): `benchmarks/2026-10-05-followup/results/calibration_profile_free_tier_vpc_dev15_benchmark.json`
- Evaluation Harness & Prompts: `benchmarks/2026-10-05-followup/harness/` and `benchmarks/2026-10-05-followup/prompts/`

1. Executive Summary: What Changed & Why It Matters

Following our October 19 review—and consistent with our September 28 agreement to evaluate candidate open-weight models over hosted APIs before deploying to target project hardware—we expanded our model evaluation in two directions:

1. **Evaluating Open-Weight Models Sized for Target Self-Hosted GPU Tiers (Informs Client Question `Q3` / `D51`):** Rather than restricting evaluation to 8B–9B models that fit on a 16 GB laptop, we organized open-weight candidates (all Apache 2.0, MIT, Llama, or Gemma licensed for self-hosting with zero external data egress) by the **GPU memory required when self-hosted**: **`Tier 0: <1 GB` (no LLM)**, **`Tier 1: 1x 24–48 GB GPU` (`L4` / `L40S`)**, **`Tier 2: 1x 80 GB GPU` (`A100` / `H100`)**, and **`Tier 3: Multi-GPU 160 GB+`**.
2. **Combining Token-Level Zero-Shot Span Detection (`GLiNER`) with Multi-Architecture LLM Verification (`Configs A, B, C, D`):** We integrated a lightweight `300M`-parameter bidirectional span encoder (`urchade/gliner_multi_pii-v1`) ahead of our LLM stages, and designed a proposed **Multi-Document Evaluation Suite** ready to test non-email document types if confirmed by Virtual Gold (**Client Question `Q4` / `A15`**).

> **Scope & Measurement Qualifications:**
>
> - **Execution Environment:** All 9 new configurations were evaluated over **hosted open-weight model APIs**; the service is not yet deployed in a client VPC or GPU sandbox (**see Question `Q3`**). Reported GPU sizes describe what each model requires when self-hosted, and reported speeds (`1.6s–2.8s`) reflect hosted API execution rather than client hardware.
> - **Dataset Scope:** Evaluated on the same **15 development emails (`DEFAULT_15_IDS`)** used on October 19, covering **4 of the 15 business domains** (`Finance & Accounting`, `HR & People`, `Legal & Compliance`, `Sales & Customers`). Earlier `Llama 3.1 8B` and `Qwen 3.5 9B` comparison rows are prior local Apple M4 runs from the October 19 harness.
> - **Tier Policy Alignment:** Figures below were measured under the benchmark harness's 3-tier rules (`HIGH ≥ 0.20`, `MEDIUM ≥ 0.50`, `LOW ≥ 0.50`). Once Virtual Gold confirms the updated tier table and acceptable leak targets (**Questions `Q1` and `Q2`**), we will re-score all configurations directly against `service/config/policy.yaml` and extend the run from 15 to 60 development emails.

Headline Achievements in This Update:

1. **Audited & Patched 6 Pre-Existing Harness & Calibration Defects (Section 2):**
    - Resolved the **Stage 2 Unmasking Deadlock** (`0/133` candidates unmasked in prior runs due to an unfitted bin 0 prior of `0.60 > 0.55` Medium threshold), restoring active Stage 2 unmasking of benign corporate context (`Mode C—Raw`).
    - Closed **omitted-candidate and ambiguous-score fail-open leaks** in `harness/pipeline.py` so any candidate dropped by Stage 2 or scored in the ambiguous zone (`0.35–0.65`) automatically **fails closed (`PRIVATE_MASKED`)**.
    - Fixed **High-Tier Deterministic Locking** (`has_high_tier_rule_hit`) so `HIGH`-tier regex hits (`SSN`, `CREDIT_CARD`, `BANK_ROUTING`, `PASSWORD`, `API_KEY`, `CRYPTO_KEY`) never lose their lock during span merging, while LLM-proposed `FACT` clauses remain verifiable in Stage 2.

- Aligned `prompts/assessment_prompt_v2.txt` with the benchmark gold policy so **confidential M&A targets, transaction counterparties, unnamed strategic partners, and outside legal counsel (`ORG` / `CODENAME`)** are protected as `PRIVATE` rather than unmasked as "published corporate names."
- Eliminated **Stage 2 output token bloat** by removing the unused per-candidate `"reasoning"` field from the Stage 2 JSON schema, reducing Stage 2 completion tokens by **~75%** and cutting Stage 2 API latency to **`0.65s–0.98s`**.
- Eliminated **silent cross-model fallback** (`allow_local_fallback=False`) and added strict provider/quantization pinning (`VPC_MODEL_REGISTRY`) in `harness/model_client.py`.

2. **Integrated `GLiNER` (`300M`) Zero-Shot Span Encoder & 4 Pipeline Architectures (Section 3):**
   - **`Baseline–B0` (`Rules + Presidio + GLiNER`, No LLM Calls):** Adding `urchade/gliner_multi_pii-v1` (`300M` DeBERTa-v3 bidirectional span encoder, `<1 GB` memory) with 220-word sliding-window chunking boosted discrete entity recall from **`77.25%` to `90.42%` (+13.17 percentage points)**, raised `PERSON` recall from `82.09%` to **`97.01%`**, raised `ORG` recall from `9.09%` to **`63.64%`**, and cut private character leakage from `42.82%` to **`29.39%`** in **`0.11s` median latency** without calling any LLM.
3. **Evaluated 9 New Configurations on the 15-Email Benchmark (`135` Email Runs, Section 4):**
   - **Best Privacy Protection (`Config B`: `Qwen 3.8 27B + GLiNER`, sized for one 24–48 GB GPU):**
     - Achieved **`7.44%` Private Character Leakage (`226 / 3,038` chars)** — **halving the October 19 `Llama 3.3 70B` leak rate (`16.19%`, `492 / 3,038` chars)** and reducing leakage by **77% compared to local `Llama 3.1 8B` (`32.36%`)**, at the cost of higher public text over-masking (`21.72%`).
     - Achieved **`11.44%` `FACT` Character Leakage (`96 / 839` chars)** and **`90.00%` `FACT` Clause Recall (`9 / 10` facts)** — compared to `60.00%` (`6 / 10` facts) and `36.95%` `FACT` leakage on `Llama 3.3 70B`.
     - Achieved **`92.81%` Exact Entity Recall (`94.01%` partial recall)** and **`98.51%` `PERSON` recall (`66 / 67`)** at **`2.37s` median hosted API latency** with **`0` operational failures**.
   - **Lowest Public Over-Masking (`Config A` & `Config C`: `OpenAI GPT-OSS 120B MoE`, sized for one 80 GB GPU):**
     - In Two-Stage mode (`Config A`), achieved **`11.54%` Public Character Over-Masking** and only **`1` extra false-positive mask** across all 15 emails (`2.30s` median hosted API latency), but leaked more private characters (`22.94%`).
     - In Single-Pass mode (`Config C`: `gpt-oss-120b+1pass`), achieved **`5.44%` Public Character Over-Masking**, **`89.82%` Entity Recall**, and **`1.62s` median hosted API latency** (`26.99%` private character leakage).
   - **Asymmetric Two-Model Checker (`Config D`: `Qwen 3.8 27B` Stage 1 + `OpenAI GPT-OSS Safeguard 20B` Stage 2, sized for one 24–48 GB GPU):**
     - Routing Stage 2 verification to OpenAI's policy-specialized `gpt-oss-safeguard-20b` (`20B` MoE, `3.6B` active parameters) reduced public over-masking by **3.60 percentage points** (`18.27%` → `14.67%`), reduced extra false-positive masks to **`3`**, caught **`8 of 10` confidential facts**, and executed in **`1.75s` median hosted API latency**.

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2. Six Harness & Calibration Defects Resolved

Before running the new configurations, we audited the evaluation harness (`harness/model_client.py`, `harness/pipeline.py`, `harness/detectors.py`, `harness/calibration.py`, `harness/run_expanded_dev_benchmark.py`, and `prompts/assessment_prompt_v2.txt`) and resolved **six pre-existing defects** that had previously affected Stage 2 behavior and cross-model comparisons:

2.1 Defect #1: Silent Cross-Model Fallback to Local `llama3.1:8b` (`harness/model_client.py`)

- **Root Cause:** In `harness/model_client.py`, `call_model()` defaulted to `allow_local_fallback=True`. Whenever a cloud endpoint experienced a transient rate limit (HTTP 429) or timeout, the client silently invoked local `llama3.1:8b` via Ollama and recorded the output under the cloud model's trace.
- **Engineering Fix:**
  1. Set `allow_local_fallback: bool = False` by default across all benchmark runs.
  2. Added exponential backoff with jitter (`5` retries) for HTTP 429/502/503/504 and automatic Tokens-Per-Minute (TPM) pacing for Groq endpoints (`_call_groq()`).

3. Added a strict provider-pinning registry (`VPC_MODEL_REGISTRY`) locking `provider_order`, `allow_fallbacks: False`, quantization (`fp8` / `bf16` / `mxfp4`), and reasoning suppression (`reasoning: {"effort": "none"}` or `"low"`) per model.

2.2 Defect #2: Stage 2 Unmasking Deadlock (`0 / 133` Unmasked in Prior Runs) (`harness/pipeline.py` & `harness/calibration.py`)

- **Root Cause:** Previously, `harness/pipeline.py` loaded `config/calibration_profile_v1.json` at import time. In that profile, bin 0 (`raw_score` $\in [0.0, 0.2)$) had an empirical value of `0.6000` (fitted from an earlier uncalibrated run where deterministic overrides and prompt-mismatched `ORG`s were mixed into bin 0). Because `0.6000` exceeded the `MEDIUM` (`0.50`) and `LOW` (`0.50`) tier thresholds, **even when a Stage 2 LLM classified a candidate as `PUBLIC` with `1.00` confidence (`raw_p_private = 0.00`), the calibrator mapped its score to `0.6000 >= 0.50` and forced it to remain `REDACTED`**. In `local_m4_dev15_benchmark.json`, `0` out of `133` non-locked candidates were unmasked by Stage 2 (`Mode B == Mode C`).
- **Engineering Fix:**
  1. Changed `PipelineConfig` default to `use_calibrated_scores: bool = False` (`Mode C-Raw`), evaluating Stage 2 on direct raw privacy probability (p_{priv}) during model selection, while reporting calibrated scores (`Mode C-Calibrated`) as an explicit post-hoc analysis layer.
  2. Fixed the fallback prior in `config/calibration_profile_v1.json` (`bin 0 = 0.10`, `is_fitted = false`).
  3. Updated `harness/calibration.py` to exclude all policy overrides (`deterministic_policy_override`, `omitted_fail_closed_override`, `ambiguous_fail_closed_override`) before fitting empirical bins, and enforced monotonic non-decreasing bin probabilities.

2.3 Defect #3: Omitted-Candidate & Ambiguous-Score Fail-Open Leaks (`harness/pipeline.py`)

- **Root Cause:** In `harness/pipeline.py`, if the Stage 2 LLM omitted a candidate ID from its returned JSON array, `decision_map.get(cid)` defaulted to `{"classification": "PUBLIC", "confidence": 0.5}`, mapping to `raw_p_private = 0.5` —and if `detector == "model_detection"`, the multi-detector ensemble produced `0.30 < 0.50`, **unmasking the omitted span (failing open)**.
- **Engineering Fix:** Patched `_evaluate_candidate_decisions()` so that any candidate omitted by Stage 2 (`omitted_by_stage2 = cid not in decision_map`) or returned with `classification == "UNKNOWN"` or ambiguous confidence (`0.35 <= raw_p_private <= 0.65`) **unconditionally fails closed (`should_redact = True`, `decision = "REDACT"`)** when the candidate is High/Medium tier or identifying context is present.

2.4 Defect #4: Deterministic High-Tier Lock Bypass & `FACT` Over-Locking (`harness/detectors.py` & `harness/pipeline.py`)

- **Root Cause:** Two subtle bugs interacted in span merging and locking:
  1. When `merge_and_deduplicate_spans()` merged an overlapping `HIGH`-tier regex match (e.g., `SSN`) with a longer `MEDIUM`-tier span, it kept the longer span's category (`MEDIUM`), bypassing the `HIGH`-tier lock.
  2. Conversely, Stage 2 locked any span where `tier == "HIGH" and ("regex" in detector or "presidio" in detector)` —which unintentionally locked LLM-proposed `FACT` spans whenever a `FACT` clause happened to absorb a single regex token (like a date or dollar amount) during merging.
- **Engineering Fix:**
  1. Added strict category precedence (`FACT > HIGH > MEDIUM > LOW`) in `harness/detectors.py` along with an explicit `normalize_category()` mapping all 18 canonical gold dataset categories.
  2. Added a boolean flag `has_high_tier_rule_hit` computed on each individual regex/Presidio span *before* merging and preserved via logical OR (`|`) across merged spans. Stage 2 now applies `deterministic_policy_override` strictly when `s.get("has_high_tier_rule_hit", False)` is `True`.

2.5 Defect #5: Confidential Counterparty `ORG` Prompt Policy Mismatch (`prompts/assessment_prompt_v2.txt`)

- **Root Cause:** In the gold benchmark dataset (`baseline_dev.jsonl`), acquisition targets, customer accounts under negotiation, and external law firms (`Castell Data Systems`, `Keane Porter LLP`, `Greenvale Grocers`, `Hartwell Department Stores`) are labeled `must_mask` (`ORG`). However, `prompts/assessment_prompt_v2.txt` previously instructed Stage 2 that "*Published corporate entities, vendor names, and official titles are PUBLIC,*" causing Stage 2 models to systematically unmask confidential counterparties.
- **Engineering Fix:** Updated `prompts/assessment_prompt_v2.txt` so `"PRIVATE"` explicitly includes "*confidential transaction counterparties, acquisition targets, unnamed partners, or legal counsel (`ORG` / `CODENAME`),*" while `"PUBLIC"` is restricted to "*general non-sensitive corporate references (e.g., public vendor names or standard software tools not tied to a*

confidential deal, dispute, or client account)."

2.6 Defect #6: Stage 2 Per-Candidate `"reasoning"` Token Bloat (`prompts/assessment_prompt_v2.txt` & `harness/model_client.py`)

- **Root Cause:** Requiring Stage 2 to generate a free-text `"reasoning"` string for every candidate inflated Stage 2 output length by `300–1,200` tokens per email (and caused truncation on documents with `20+` candidates), even though `_evaluate_candidate_decisions()` never read the `"reasoning"` field.
- **Engineering Fix:** Removed `"reasoning"` from `prompts/assessment_prompt_v2.txt` and `STAGE2_ASSESSMENT_JSON_SCHEMA` in `harness/model_client.py`, reducing Stage 2 completion length to `~120–160` tokens and cutting Stage 2 API latency to **`0.65s–0.98s`**.

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3. Target Self-Hosted GPU Tiers & Four Pipeline Architectures

3.1 Candidate Open-Weight Models Sized by Self-Hosted GPU Footprint

We organized the 13 open-weight models in our evaluation registry (`VPC_MODEL_REGISTRY` in `harness/model_client.py`) by the GPU memory footprint each model requires when self-hosted (via vLLM, SGLang, or Ollama):

Target GPU Tier (When Self-Hosted)	Example Cloud / On-Prem GPU	VRAM Budget	Registered Open-Weight Models	License (Self-Hostable)	Target Role
Tier 0: Encoder Companion	CPU or `<1 GB` shared GPU	`<1 GB`	**`urchade/gliner_multi_pii-v1`** (`300M` DeBERTa-v3 bi-encoder)	Apache 2.0	Fast (`0.11s`) token-level zero-shot PII & entity span pre-detector (`Config B` & degraded fallback)
Tier 1: Single 24–48 GB GPU	`1x NVIDIA L4` (24GB) or `1x L40S` (48GB)	`24–48 GB`	**`qwen/qwen3.8-27b`** (27B Dense) **`openai/gpt-oss-safeguard-20b`** (20B MoE, 3.6B act.) `mistralai/mistral-small-3.2-24b-instruct` `google/gemma-4-26b-a4b-it` (26B MoE, 4B act.) `qwen/qwen3.6-35b-a3b` (35B MoE, 3B act.) `qwen/qwen3-32b` (32B Dense)	Apache 2.0 / Gemma	**Primary cost-effective single-GPU deployment tier**
Tier 2: Single 80 GB GPU	`1x NVIDIA H100` or `A100` (80GB)	`80 GB`	**`openai/gpt-oss-120b`** (117B MoE, 5.1B act., native MXFP4) `meta-llama/llama-3.3-70b-instruct` (70B Dense) `meta-llama/llama-4-scout-17b-16e-instruct` (109B MoE) `qwen/qwen3.5-122b-a10b` (122B MoE, 10B act.)	Apache 2.0 / Llama	**High-precision single-80GB node tier**
Tier 3: Multi-GPU Ceiling	`2x–8x H100 / H200` (160–640GB)	`160 GB+`	`qwen/qwen3-235b-a22b-instruct-2507` (235B MoE, 22B act.) `deepseek/deepseek-chat-v3.2` (685B MoE, 37B act.)	Apache 2.0 / MIT	**Upper-bound accuracy ceiling benchmark**

3.2 Four Pipeline Architecture Variants (`Configs A, B, C, D`)

All four architecture variants are implemented in `harness/pipeline.py` and `harness/detectors.py`:

```
flowchart TB
    Doc["Input Enterprise Text (Executive Email; Proposed Extension to Documents – see Q4)"]

    subgraph Tier0["Stage 0: Deterministic & Zero-Shot Pre-Detectors (<120ms)"]
        Reg["Regex + Presidio + Header-to-Body Name Propagation + Clinical Cues"]
        Gli["GLiNER 300M Bi-Encoder (220-word sliding windows, Config B / Baseline-B0)"]
    end

    Doc --> Reg
    Doc --> Gli

    subgraph Arch["LLM Pipeline Architecture Variants"]
        direction TB
        CA["Config A (Symmetric 2-Stage): Stage 1 Extractor LLM → Stage 2 Verifier LLM (Same Model)"]
        CB["Config B (+GLiNER 2-Stage): Rules + Presidio + GLiNER → Stage 1 Extractor LLM → Stage 2 Verifier LLM"]
        CC["Config C (+1pass Single-Pass): Rules + Presidio → Single Merged Prompt (Baseline Adjudication + New Spans)"
        CD["Config D (Asymmetric 2-Model): Stage 1 High-Recall Extractor (Qwen 27B) → Stage 2 Policy Checker (GPT-OSS 
    end

    Reg --> Arch
    Gli -.->|"Enabled in Baseline-B0, Config B, Config B+D"| Arch

    subgraph Gate["Deterministic Safety Policy Gate"]
        Lock["1. High-Tier Rule Lock (has_high_tier_rule_hit = True → Always REDACT)"]
        FailClosed["2. Omitted / Ambiguous Candidate Fail-Closed Override (Always REDACT)"]
        Thresh["3. Tier Threshold Gate (Current Harness: HIGH ≥ 0.20, MEDIUM ≥ 0.50, LOW ≥ 0.50; Re-scoring on Confirme
    end

    Arch --> Gate
    Gate --> Out["Redacted Text ([PERSON_1], [FACT_1], ...) + Reversible Placeholder Map + Audit Telemetry"]
```

4. 15-Email Benchmark Results (12 Configurations Compared)

We evaluated **9 new configurations** (2 zero-LLM local baselines + 7 open-weight LLM configurations run via hosted APIs) on the 15-email development set (`DEFAULT_15_IDS`: `E001` – `E024`, covering **4 of 15 business domains**, containing **`3,038` private characters**, **`839` confidential `FACT` characters**, **`5,277` public characters**, **`167` discrete entity spans**, and **`10` confidential business `FACT` clauses**) and compared them against our **3 October 19 reference baselines** (`Local Llama 3.1 8B`, `Local Qwen 3.5 9B`, and `Hosted Llama 3.3 70B`):

4.1 Master End-to-End Performance Table (Stage C / Policy-Gated Output)

#	Configuration / Model ID	GPU Needed When Self-Hosted (How Tested)	Private Char Leak ↓	FACT Char Leak ↓	Public Over-Mask ↓	Exact Entity Recall ↑	Partial Entity Recall ↑	FACT Clause Recall ↑	PERSON Recall ↑	Clean Emails ↑	Extra Masks ↓	Op / Schema Fail	Median Seconds / Email (S1 / S2)
1	**Baseline-A (Rules + Presidio)**	None (Local CPU, 0 LLM)	42.82% (1301)	98.93% (830)	**2.29%** (121)	77.25%	79.64%	0.00% (0/10)	82.09%	3/15	6	0 / 0	**0.00s**
2	**Baseline-B0 (Rules + Presidio + GLiNER)**	<1 GB (Local, **0 LLM**)	**29.39%** (893)	84.03% (705)	6.58% (347)	**90.42%**	**92.81%**	0.00% (0/10)	**97.01%**	**7/15**	14	0 / 0	**0.11s**
3	*Oct 19 Local Ref:* Llama 3.1 8B *	16 GB RAM (Local M4, 2-Stage)	32.36% (983)	70.92% (595)	9.72% (513)	77.25%	—	20.00% (2/10)	64.18%	1/15	17	0 / 0	77.60s (24.3s / 52.9s)
4	*Oct 19 Local Ref:* Qwen 3.5 9B *	16 GB RAM (Local M4, 2-Stage)	38.22% (1161)	68.06% (571)	6.05% (319)	70.66%	—	40.00% (4/10)	55.22%	2/15	3	0 / 0	110.60s (37.5s / 78.1s)
5	*Oct 19 Ref:* Llama 3.3 70B *	80 GB GPU (Hosted API, 2-Stage)	16.19% (492)	36.95% (310)	18.46% (974)	**92.81%**	—	60.00% (6/10)	**100.0%**	**7/15**	7	0 / 1	40.84s (15.6s / 20.4s)
6	**Qwen 3.8 27B (Config A)**	**24–48 GB GPU** (Hosted API, 2-Stage)	**11.72%** (356)	**14.54%** (122)	18.27% (964)	90.42%	91.02%	**90.00%** (9/10)	**98.51%**	6/15	4	**0 / 0**	**1.79s** (0.90s / 0.84s)
7	**Qwen 3.8 27B + GLiNER (Config B)** 🏆	**24–48 GB GPU** (Hosted API, 2-Stage)	**7.44%** (226)	**11.44%** (96)	21.72% (1146)	**92.81%**	**94.01%**	**90.00%** (9/10)	**98.51%**	**7/15**	8	**0 / 0**	**2.37s** (0.92s / 0.98s)
8	**Qwen 3.8 27B :: Safeguard 20B (Config D)**	**24–48 GB GPU** (Hosted API, S1 27B + S2 20B)	13.96% (424)	23.12% (194)	**14.67%** (774)	89.22%	89.82%	80.00% (8/10)	97.01%	5/15	**3**	**0 / 0**	**1.75s** (0.94s / 0.65s)
9	**Qwen 27B + GLiNER :: Safeguard 20B (B+D)**	**24–48 GB GPU** (Hosted API, S1 27B+ GLiNER + S2 20B)	12.67% (385)	25.86% (217)	17.49% (923)	90.42%	91.62%	70.00% (7/10)	97.01%	5/15	7	**0 / 0**	**2.08s** (0.96s / 0.68s)
10	**OpenAI GPT-OSS 120B (Config A)** 🎯	**80 GB GPU** (Hosted API, 2-Stage)	22.94% (697)	52.80% (443)	**11.54%** (609)	88.02%	88.62%	40.00% (4/10)	97.01%	4/15	**1**	0 / 1 **	**2.30s** (1.16s / 1.14s)
11	**OpenAI GPT-OSS 120B + GLiNER (Config B)**	**80 GB GPU** (Hosted API, 2-Stage)	17.25% (524)	43.50% (365)	12.81% (676)	**91.02%**	**92.22%**	50.00% (5/10)	97.01%	6/15	7	0 / 2 **	**2.85s** (1.22s / 1.29s)
12	**OpenAI GPT-OSS 120B (Config C: 1-Pass)** ⚡	**80 GB GPU** (Hosted API, Single-Pass)	26.99% (820)	73.54% (617)	**5.44%** (287)	89.82%	91.62%	30.00% (3/10)	95.52%	4/15	12	**0 / 0**	**1.62s** (1.62s / 0.00s)

**Rows 3–5 are from the October 19 evaluation prior to the six harness fixes in Section 2; Rows 3–4 ran locally via Ollama on an Apple Silicon M4 laptop (16 GB RAM).*

On the hosted endpoint (`1,000 RPM / 200K TPD`), `gpt-oss-120b` experienced `1` JSON truncation in `Config A` and `2` in `Config B` when internal reasoning tokens consumed part of the `2,048` completion token budget on dense candidate lists; the pipeline safely fell back to deterministic + Stage 1 spans. On successful calls only, `gpt-oss-120b + GLiNER` achieved* ***`14.83%` private character

***leakage** and `13.89%` **public over-masking**.*

4.2 Entity Recall Breakdown by Category (All 14 Canonical Entity Types)

Below is the exact entity recall (`score.py` ≥ 50% token coverage criterion) across all **14 discrete entity categories** (`167` spans total) and `FACT` **clauses** (`10` spans total) on the 15-email benchmark:

Entity Category (Count)	Baseline–A (Rules Only)	Baseline–B0 (+GLiNER, 0 LLM)	Oct 19 Llama 3.3 70B (Config A)	Qwen 3.8 27B (Config A)	Qwen 3.8 27B + GLiNER (Config B) 🏆	Qwen 27B :: Safeguard 20B (Config D)	GPT–OSS 120B (Config A)	GPT–OSS 120B + GLiNER (Config B)	GPT–OSS 120B (Config C: 1–Pass)
`ADDRESS (1)`	0.00% (`0/1`)	0.00% (`0/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)
`AMOUNT (31)`	93.55% (`29/31`)	**100.0%** (`31/31`)	**100.0%** (`31/31`)	**100.0%** (`31/31`)	**100.0%** (`31/31`)	**100.0%** (`31/31`)	**100.0%** (`31/31`)	**100.0%** (`31/31`)	93.55% (`29/31`)
`CODENAME (2)`	0.00% (`0/2`)	50.00% (`1/2`)	50.00% (`1/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)	0.00% (`0/2`)	50.00% (`1/2`)	0.00% (`0/2`)
`CREDENTIAL (1)`	0.00% (`0/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)
`DOB (1)`	0.00% (`0/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)
`EMAIL (32)`	**100.0%** (`32/32`)	**100.0%** (`32/32`)	93.75% (`30/32`)	93.75% (`30/32`)	**100.0%** (`32/32`)	96.88% (`31/32`)	96.88% (`31/32`)	**100.0%** (`32/32`)	96.88% (`31/32`)
`FIN_ACCOUNT (4)`	**100.0%** (`4/4`)	**100.0%** (`4/4`)	**100.0%** (`4/4`)	**100.0%** (`4/4`)	**100.0%** (`4/4`)	**100.0%** (`4/4`)	**100.0%** (`4/4`)	**100.0%** (`4/4`)	**100.0%** (`4/4`)
`GOV_ID (2)`	50.00% (`1/2`)	50.00% (`1/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)	**100.0%** (`2/2`)
`HEALTH (2)`	0.00% (`0/2`)	0.00% (`0/2`)	**50.00%** (`1/2`)	**50.00%** (`1/2`)	**50.00%** (`1/2`)	**50.00%** (`1/2`)	**50.00%** (`1/2`)	**50.00%** (`1/2`)	0.00% (`0/2`)
`ID_NUMBER (6)`	16.67% (`1/6`)	16.67% (`1/6`)	**66.67%** (`4/6`)	33.33% (`2/6`)	33.33% (`2/6`)	16.67% (`1/6`)	16.67% (`1/6`)	16.67% (`1/6`)	50.00% (`3/6`)
`LOCATION (1)`	0.00% (`0/1`)	**100.0%** (`1/1`)	0.00% (`0/1`)	0.00% (`0/1`)	**100.0%** (`1/1`)	0.00% (`0/1`)	0.00% (`0/1`)	**100.0%** (`1/1`)	**100.0%** (`1/1`)
`ORG (11)`	9.09% (`1/11`)	**63.64%** (`7/11`)	54.55% (`6/11`)	36.36% (`4/11`)	45.45% (`5/11`)	27.27% (`3/11`)	27.27% (`3/11`)	45.45% (`5/11`)	**63.64%** (`7/11`)
`PERSON (67)`	82.09% (`55/67`)	97.01% (`65/67`)	**100.0%** (`67/67`)	**98.51%** (`66/67`)	**98.51%** (`66/67`)	97.01% (`65/67`)	97.01% (`65/67`)	97.01% (`65/67`)	95.52% (`64/67`)
`PHONE (6)`	**100.0%** (`6/6`)	**100.0%** (`6/6`)	**100.0%** (`6/6`)	**100.0%** (`6/6`)	**100.0%** (`6/6`)	**100.0%** (`6/6`)	**100.0%** (`6/6`)	**100.0%** (`6/6`)	**100.0%** (`6/6`)
Total Entities (167)	**77.25%**	**90.42%**	**92.81%**	**90.42%**	**92.81%** (`94.01%` part.)	**89.22%**	**88.02%**	**91.02%** (`92.22%` part.)	**89.82%**
`FACT` Clauses (10)	0.00% (`0/10`)	0.00% (`0/10`)	60.00% (`6/10`)	**90.00%** (`9/10`)	**90.00%** (`9/10`)	**80.00%** (`8/10`)	40.00% (`4/10`)	50.00% (`5/10`)	30.00% (`3/10`)

4.3 Entity Recall Breakdown by Enterprise Domain (4 of 15 Domains Represented in Dev-15)

Across the four functional enterprise domains present in the 15-email subset (`Finance & Accounting`, `HR & People`, `Legal & Compliance`, `Sales & Customers`), `Qwen 3.8 27B + GLiNER (Config B)` achieved the highest or tied-highest recall in every category:

Enterprise Business Domain	Baseline-A (Rules Only)	Baseline-B0 (+GLiNER, 0 LLM)	Qwen 3.8 27B (Config A)	Qwen 3.8 27B + GLiNER (Config B) 🏆	Qwen 27B :: Safeguard 20B (Config D)	GPT-OSS 120B (Config A)	GPT-OSS 120B + GLiNER (Config B)	GPT-OSS 120B (Config C: 1-Pass)
Finance & Accounting	90.91%	**98.48%**	95.45%	**98.48%**	95.45%	95.45%	**98.48%**	95.45%
HR & People	69.77%	90.70%	93.02%	**95.35%**	93.02%	86.05%	93.02%	88.37%
Legal & Compliance	61.70%	76.60%	80.85%	**82.98%**	76.60%	78.72%	78.72%	**82.98%**
Sales & Customers	90.91%	**100.00%**	90.91%	90.91%	90.91%	90.91%	90.91%	90.91%

5. Deep-Dive Analysis of Key Findings

5.1 Why `Qwen 3.8 27B + GLiNER` (`Config B`, 24–48 GB GPU Tier) Outperforms `Llama 3.3 70B` (`Config A`, 80 GB GPU Tier)

In our October 19 report, `Llama 3.3 70B` was our top enterprise candidate (`16.19%` private character leakage, `60.0%` `FACT` recall, `92.81%` entity recall, `40.84s` median latency). Our new Tier 1 configuration— **`Qwen 3.8 27B + GLiNER` (`Config B`)**—substantially improves privacy recall while requiring **half the GPU VRAM (`24–48 GB` vs `80 GB`)**:

1. **54% Lower Private Character Leakage:** `7.44%` (`226 / 3,038` chars) vs. `16.19%` (`492 / 3,038` chars) on `Llama 3.3 70B`.
2. **69% Lower Confidential `FACT` Character Leakage:** `11.44%` (`96 / 839` chars) vs. `36.95%` (`310 / 839` chars) on `Llama 3.3 70B`.
3. **90% vs. 60% Confidential `FACT` Clause Recall:** `Qwen 3.8 27B` caught **9 out of 10** complex multi-clause confidential business facts (and achieved `95%` token coverage on the remaining medical `HEALTH` clause in `E018`).
4. **100% Project Codename Recall:** Caught **100% (`2/2`) of `CODENAME` entities** (vs. `50%` on `Llama 3.3 70B`).
5. **Tradeoff:** Its aggressive recall masks slightly more public text (`21.72%` vs. `18.46%` on `Llama 3.3 70B` and `11.54%` on `GPT-OSS 120B`).

5.2 Why `GLiNER` (`300M`) Is a Strong Addition for Stage 1 Span Recall & Degraded Fallback

Autoregressive LLMs generate text token-by-token and occasionally miss secondary mentions of names, locations, or organizations in dense paragraphs. In contrast, **`GLiNER` (`urchade/gliner_multi_pii-v1`)** is a bidirectional DeBERTa-v3 span encoder (`<1 GB` memory, Apache 2.0 license) that scores start/end token pairs directly:

- **Zero-LLM Degraded Fallback (`Baseline-B0`):** Even with **zero LLM calls**, adding `GLiNER` to Regex + Presidio jumps overall entity recall from **`77.25%` to `90.42%` (+13.17 pp)**, catches **`97.01%` (`65/67`) of `PERSON` names**, **`100%` of `CREDENTIAL`, `DOB`, `LOCATION`, and `AMOUNT` spans**, and **`63.64%` (`7/11`) of `ORG` spans** in **`0.11s`**. This makes `Rules + Presidio + GLiNER` a strong candidate for the service's degraded-mode fallback when the LLM is unavailable.
- **Hybrid LLM Mode (`Config B`):** When `GLiNER` pre-populates structured entity candidates, the Stage 1 LLM no longer misses `EMAIL` (`93.75%` → **`100.0%`**) or `LOCATION` (`0.0%` → **`100.0%`**) and can dedicate its attention to extracting full-clause `FACT` statements (`90.0%` recall).

5.3 Operating Frontier: Choosing Between `Qwen 3.8 27B`, `Safeguard 20B` (`Config D`), and `GPT-OSS 120B`

Depending on the acceptable leak and over-masking targets Virtual Gold sets in **Question `Q2`**:

- **Lowest Private Leakage (`Qwen 3.8 27B + GLiNER`, `Config B`, 24–48 GB GPU):** Best when minimizing confidential fact and PII leakage is the primary priority (`7.44%` private char leak, `90%` `FACT` recall, `21.72%` public over-masking).
- **Balanced Two-Model Checker (`Qwen 3.8 27B :: GPT-OSS Safeguard 20B`, `Config D`, 24–48 GB GPU):** Uses `Qwen 3.8 27B` for Stage 1 extraction and `GPT-OSS Safeguard 20B` as the Stage 2 policy checker, cutting public over-masking to **`14.67%`** and extra masks to **`3`** while catching **`8 of 10` facts** (`13.96%` private char leak).
- **Lowest Public Over-Masking (`OpenAI GPT-OSS 120B`, `Config A` / `Config C`, 80 GB GPU):** Masks only **`11.54%`** (`Config A`, `1` extra mask) to **`5.44%`** (`Config C`, single-pass) of public text, but leaks more confidential facts (`22.94%`–`26.99%` private char leak) and requires an 80 GB GPU.

6. Updated Empirical Score Calibration (`v1.4.0` on `1,198` Candidate Audit Records)

Using our patched calibration engine (`harness/calibration.py`), we generated `results/calibration_profile_free_tier_vpc_dev15_benchmark.json` across all **`1,198` candidate decisions** collected in the new benchmark runs:

- **Total Audited Candidate Decisions:** `1,198`
- **Deterministic / Fail-Closed Policy Overrides Cleanly Excluded:** `55` candidates (`score_method` ∈ `{"deterministic_policy_override", "omitted_fail_closed_override", "ambiguous_fail_closed_override"}`, recorded with `calibrated_score: null`).
- **Pure Model-Evaluated Candidates Fitted:** `1,143` candidates:

Raw Score Bin	Candidate Count	Monotonic Empirical *P*(Private)	Behavior After Prompt & Lock Fixes
`[0.0, 0.2)` (`PUBLIC` Verdict)	`100`	**`0.2765`** (down from `0.6000` previously!)	**Resolved Unmasking Deadlock:** Because `0.2765 < 0.50` (`MEDIUM` / `LOW` threshold), even under calibrated scoring (`Mode C–Calibrated`), confident `PUBLIC` verdicts now properly unmask benign corporate entities!
`[0.2, 0.4)`	`1`	**`0.3000`**	Sparse low-confidence region (monotonic floor enforced)
`[0.4, 0.6)`	`0`	**`0.5000`**	Ambiguous midpoint prior (`[0.35, 0.65]` triggers fail-closed policy)
`[0.6, 0.8)`	`0`	**`0.7000`**	Moderate-confidence private prior
`[0.8, 1.01)` (`PRIVATE` Verdict)	`1,042`	**`0.9672`** (up from `0.9447` previously!)	**High Precision on `PRIVATE` Verdicts:** `96.72%` of candidates classified as `PRIVATE` by the evaluated models overlap true gold private spans.

- **Calibration Quality Metrics (`1,143` Model-Evaluated Candidates):**
  - **Raw Expected Calibration Error (ECE):** `0.0324` → **Calibrated ECE:** **`0.0011`** (`−96.6%` calibration error)
  - **Raw Brier Score:** `0.0487` → **Calibrated Brier Score:** **`0.0468`** (down from `0.1180` raw / `0.0796` calibrated in the October 19 run)

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7. Questions for Virtual Gold & Next-Phase Evaluation Roadmap

7.1 Alignment with Client Decisions Today (`Q1`–`Q4` in the Meeting Agenda)

1. **Tier Table & Acceptable Leak Targets (`Q1` & `Q2`):** Once Virtual Gold confirms the updated tier table (`policy 0.1.3` vs. `0.1.1`) and acceptable leak/over-masking rates per tier, we will re-score our benchmark directly against `service/config/policy.yaml` and extend evaluation from **15 to 60 development emails**.
2. **Target Deployment Environment (`Q3` / `D51`):** Our best configuration (`Qwen 3.8 27B + GLiNER`) fits on a single **24–48 GB GPU**, while `GPT–OSS 120B` requires an **80 GB GPU**. Confirming where the service will run—and when a private cloud / GPU sandbox will be available—will allow us to benchmark true self-hosted latency and throughput.
3. **Document Types in Scope for Phase 1 (`Q4` / `A15`):** The SOW and midterm deck mention "*English text and documents,*" while the service currently accepts executive emails. If Virtual Gold wants non-email documents tested in Phase 1, we have designed a **Multi-Document Evaluation Suite** covering contracts, financial tables, strategy memos, HR/clinical notes, technical logs, and long transcripts (`L–Chunk` 2K sliding windows vs. `L–Native` 16K context).

7.2 Next Two Weeks for the Model Workstream

1. **Wire Prompts into the Service (`M2`, due Nov 1):** Connect the Stage 1 detection and Stage 2 assessment prompts and parsers into the HTTP service for the end-to-end pipeline milestone.
2. **Re-Score Under the Confirmed Tier Policy & Expand to 60 Dev Emails:** Apply the confirmed tier table from `Q1` / `Q2` across all 60 development emails (`baseline_dev.jsonl`).
3. **Complete Remaining Open-Weight Model Sweep:** Evaluate the remaining registered Tier 1 (`Mistral Small 3.2 24B`, `Gemma 4 26B–A4B`, `Qwen 3.6 35B–A3B`, `Qwen 3 32B`) and Tier 2 (`Llama 3.3 70B` with `GLiNER`, `Llama 4 Scout`, `Qwen 3.5 122B`) candidates.
4. **Test API v2 Long Token Preservation:** Test whether real models preserve Zhexuan's API v2 tokens (`[VG:2:…]`) without corruption during downstream processing.

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8. Updated Proposed Service Configuration (`API Contract 0.4.2`)

Below is the proposed model and policy gate configuration for the service engineering hand-off, supporting a **24–48 GB GPU profile (`Qwen 3.8 27B + GLiNER`)**, an **80 GB GPU low-overmasking profile (`GPT-OSS 120B`)**, and a **no-LLM degraded fallback (`Rules + Presidio + GLiNER`)**:

```
{
  "runtime_engine": {
    "single_gpu_24_48gb_recommended": {
      "provider": "Self-hosted vLLM / SGLang / Ollama",
      "architecture": "Config B (Rules + Presidio + GLiNER 300M -> Qwen 3.8 27B 2-Stage)",
      "stage0_encoder": "urchade/gliner_multi_pii-v1",
      "stage1_model": "qwen/qwen3.8-27b",
      "stage2_model": "qwen/qwen3.8-27b (or openai/gpt-oss-safeguard-20b for Config D)",
      "runtime_options": {
        "temperature": 0.0,
        "stage1_max_tokens": 2048,
        "stage2_max_tokens": 2048,
        "reasoning_effort": "none",
        "gpu_requirement": "1x 24-48 GB GPU (e.g., NVIDIA L4 or L40S)"
      }
    },
    "single_gpu_80gb_low_overmasking": {
      "provider": "Self-hosted vLLM / SGLang",
      "architecture": "Config B (Rules + Presidio + GLiNER 300M -> OpenAI GPT-OSS 120B MoE)",
      "stage0_encoder": "urchade/gliner_multi_pii-v1",
      "stage1_model": "openai/gpt-oss-120b",
      "stage2_model": "openai/gpt-oss-120b",
      "runtime_options": {
        "temperature": 0.0,
        "quantization": "MXFP4 (native)",
        "gpu_requirement": "1x 80 GB GPU (e.g., NVIDIA A100 or H100)"
      }
    },
    "zero_llm_degraded_fallback": {
      "architecture": "Baseline-B0 (Regex + Presidio + GLiNER 300M Only)",
      "median_latency_s": 0.11,
      "entity_recall_dev15": 0.9042,
      "person_recall_dev15": 0.9701
    }
  },
  "prompts": {
    "detection": "prompts/detection_prompt_v2.txt",
    "assessment": "prompts/assessment_prompt_v2.txt",
    "single_pass": "prompts/single_pass_prompt_v2.txt"
  },
  "policy_gate": {
    "default_gating_mode": "Mode C-Raw (use_calibrated_scores: false)",
    "high_tier_rule_lock_field": "has_high_tier_rule_hit",
    "omitted_candidate_policy": "FAIL_CLOSED_REDACT",
    "ambiguous_score_margin": [0.35, 0.65],
    "ambiguous_score_policy": "FAIL_CLOSED_REDACT",
    "tier_thresholds_note": "To be aligned with confirmed service/config/policy.yaml after Q1/Q2"
  }
}
```